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RF IMMUNE SENSOR PROBE FOR MONITORING A TEMPERATURE OF AN ELECTROSTATIC CHUCK OF A SUBSTRATE PROCESSING SYSTEM

Abstract

A sensor probe includes an elongated body defining an inner cavity having an inner diameter. A printed circuit board is configured to be fitted within the inner cavity. A first temperature-sensing integrated circuit mounted at a first end of the printed circuit board. A cap is mounted to a first end of the elongated body adjacent to the first temperature-sensing integrated circuit. A housing is configured to receive a second end of the elongated body, wherein the housing is configured to be mounted to a baseplate of a substrate support.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present disclosure is a continuation of U.S. patent application Ser. No. 17/614,930, filed on Nov. 29, 2021, which is a U.S. National Phase Application under 35 U.S.C. 371 of International Application No. PCT/US2020/035041, filed on May 29, 2020, which claims benefit of U.S. Patent Application No. 62/854,476 filed on May 30, 2019. The entire disclosures of the applications referenced above is incorporated herein by reference.

FIELD
[0002] The present disclosure relates generally to substrate processing systems and more particularly to a sensor probe for monitoring a temperature of an electrostatic chuck in a substrate processing system.
BACKGROUND
[0003] The background description provided herein is for the purpose of generally presenting the context of the disclosure. Work of the presently named inventors, to the extent it is described in this background section, as well as aspects of the description that may not otherwise qualify as prior art at the time of filing, are neither expressly nor impliedly admitted as prior art against the present disclosure.
[0004] Substrate processing systems perform treatments on substrates such as semiconductor wafers. Examples of substrate treatments include deposition, ashing, etching, cleaning and/or other processes. Process gas mixtures may be supplied to the processing chamber to treat the substrate. Plasma may be used to ignite the gases to enhance chemical reactions.
[0005] A substrate is arranged on a substrate support in the processing chamber during treatment. Changes in the temperature of the substrate support may affect the treatment. For example, deposition or etch rates may be affected by different temperatures at different locations of the substrate. As a result, deposition or etching may be different in the different locations. Some substrate supports include embedded temperature sensors to sense temperatures in multiple zones. In some examples, each of the zones includes one or more redundant temperature sensors that are used as backups when the temperature sensor for the zone fails. When all of the temperature sensors in one of the zones fail, the substrate support needs to be replaced, which can be expensive.
SUMMARY
[0006] A sensor probe includes an elongated body defining an inner cavity having an inner diameter. A printed circuit board is configured to be fitted within the inner cavity. A first temperature-sensing integrated circuit mounted at a first end of the printed circuit board. A cap is mounted to a first end of the elongated body adjacent to the first temperature-sensing integrated circuit. A housing is configured to receive a second end of the elongated body. The housing is configured to be mounted to a baseplate of a substrate support.
[0007] In other features, the printed circuit board has a width that is less than the inner diameter and a length that is longer than the elongated body. The inner diameter is less than or equal to 3 mm and at least two of three orthogonal dimensions of the first temperature-sensing integrated circuit are less than 3 mm.

[0008] In other features, potting material connects the cap and the first temperature-sensing integrated circuit. The printed circuit board is flexible and is bent at an angle adjacent to the first temperature-sensing integrated circuit. The cap includes first and second legs that extend from one side of the cap and that are received in the inner cavity of the elongated body. The elongated body is reciprocally received in the housing. The elongated body includes a projection and further comprising a spring located around the elongated body and biased between an inner cavity of the housing and the projection.
[0009] In other features, the first temperature-sensing integrated circuit senses a temperature of a surface in contact with the cap. The surface is a layer within an electrostatic chuck.
[0010] In other features, the elongated body includes a radial projection to center the elongated body in a cavity of the baseplate. The elongated body includes a slot. The slot has an elongated elliptical shape and is aligned in an axial direction of the elongated body.
[0011] In other features, a shielding layer is arranged on at least one surface of the printed circuit board. The housing defines an inclined surface. An O-ring is arranged against the inclined surface between the housing and a cavity of the baseplate.
[0012] In other features, the printed circuit board is flexible. A connector is connected to a second end of the printed circuit board. A plurality of wires is connected by the connector to traces on the printed circuit board. A second temperature-sensing integrated circuit is mounted on the printed circuit board between the first temperature-sensing integrated circuit and a second end of the printed circuit board.
[0013] A sensor probe includes an elongated body defining an inner cavity having an inner diameter. A first printed circuit board is configured to be fitted within the inner cavity. A temperature-sensing integrated circuit mounted on the first printed circuit board. A housing is configured to receive one end of the elongated body and configured to be mounted to a baseplate of a substrate support. Aa second printed circuit board is arranged in the housing. A plurality of first conductors connect the first printed circuit board to the second printed circuit board. A plurality of second conductors configured to connect the second printed circuit board to external devices.
[0014] In other features, the first printed circuit board has a width that is less than the inner diameter and a length that is less than a length of the elongated body. The second printed circuit board has a length that is less than a length of the housing. The inner diameter is less than or equal to 3 mm and at least two of three orthogonal dimensions of the temperature-sensing integrated circuit are less than 3 mm.
[0015] In other features, potting material is located inside of the elongated body. The first printed circuit board and the temperature-sensing integrated circuit are mounted parallel to a length of the elongated body. The temperature-sensing integrated circuit senses a temperature of a surface in contact therewith. The surface is a layer within an electrostatic chuck. The elongated body includes a radial projection to center the elongated body in a cavity of the baseplate.
[0016] In other features, the elongated body includes a slot. The slot has an elongated elliptical shape and is aligned in an axial direction of the elongated body. A capacitor is connected to the first printed circuit board. A resistor is connected to the second printed circuit board. A shielding layer is arranged on a surface of the first printed circuit board.
[0017] A sensor probe includes an elongated body defining an inner cavity having an inner diameter. A temperature-sensing integrated circuit is configured to be fitted within the inner cavity. A housing is configured to receive one end of the elongated body and configured to be mounted to a baseplate of a substrate support. A plurality of conductors are configured to pass through the housing and the elongated body and to connect the temperature-sensing integrated circuit to external devices.
[0018] In other features, the inner diameter is less than or equal to 3 mm and wherein at least two of three orthogonal dimensions of the temperature-sensing integrated circuit are less than 3 mm. Potting material is located inside of the elongated body. The temperature-sensing integrated circuit

is mounted parallel to a length of the elongated body. The temperature-sensing integrated circuit is mounted perpendicular to a length of the elongated body.
[0019] In other features, the temperature-sensing integrated circuit senses a temperature of a surface in contact therewith. The surface is a layer within an electrostatic chuck. The elongated body includes a radial projection to center the elongated body in a cavity of the baseplate.
[0020] In other features, the elongated body includes a slot. The slot has an elongated elliptical shape and is aligned in an axial direction of the elongated body. A plurality of solder balls is attached to the temperature-sensing integrated circuit. The plurality of first conductors are attached to the plurality of solder balls.
[0021] Further areas of applicability of the present disclosure will become apparent from the detailed description, the claims and the drawings. The detailed description and specific examples are intended for purposes of illustration only and are not intended to limit the scope of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0022] The present disclosure will become more fully understood from the detailed description and the accompanying drawings, wherein:
[0023] FIG. **1** is a functional block diagram of an example of a capacitively coupled plasma (CCP) substrate processing system including a sensor probe according to the present disclosure;
[0024] FIG. **2** is a functional block diagram of an example of an inductively coupled plasma (ICP) substrate processing system including a sensor probe according to the present disclosure;
[0025] FIG. **3** is a side cross-sectional view of an example of a substrate support including a sensor probe according to the present disclosure;
[0026] FIG. **4**A is a side view of an example of a sensor probe according to the present disclosure;
[0027] FIG. **4**B is a side cross-sectional view of an example of a sensor probe according to the present disclosure;
[0028] FIG. **5** is a plan view of an example of a printed circuit board according to the present disclosure;
[0029] FIG. **6** is an enlarged, partial side view of an example of a body of the sensor probe according to the present disclosure;
[0030] FIG. **7** is an electrical schematic and functional block diagram of an example of a control circuit including a temperature-sensing integrated circuit, resistors and a capacitor according to the present disclosure;
[0031] FIG. **8** is a side view illustrating attachment of an example of the temperature-sensing integrated circuit to the printed circuit board according to the present disclosure;
[0032] FIGS. **9**A and **9**B are side views illustrating an example of attachment of the temperature-sensing integrated circuit and the printed circuit board to a metal cap according to the present disclosure;
[0033] FIGS. **10**A and **10**B are side views illustrating an example of potting of the temperature-sensing integrated circuit to the cap according to the present disclosure;
[0034] FIGS. **11**A and **11**B are side views illustrating an example of insertion of the printed circuit board, the temperature-sensing integrated circuit, and the cap into the body of the sensor probe according to the present disclosure;
[0035] FIG. **12** is a side view illustrating another example of the sensor probe according to the present disclosure;
[0036] FIG. **13** is a side cross-sectional view illustrating another example of the sensor probe according to the present disclosure;

[0037] FIG. **14** is a side cross-sectional view of a PCB with metal layers for EMI shielding according to the present disclosure; and
[0038] FIG. **15** is a side cross-sectional view illustrating another example of the sensor probe according to the present disclosure.
[0039] In the drawings, reference numbers may be reused to identify similar and/or identical elements.
DETAILED DESCRIPTION
[0040] The present disclosure relates to sensor probes that sense a temperature of a surface in a processing chamber of a substrate. The sensor probe includes a temperature-sensing integrated circuit. In some examples, the temperature-sensing integrated circuit is located within a body that is made of metal and connected to a reference potential such as ground. For example, the body may be grounded to the baseplate. As a result, the body of the sensor probe acts as a Faraday cage and the sensor probe is immune to RF signals such as RF bias signals, electrode signals, etc. that are present in the temperature sensing environment. Alternately, the body may be made of metallic or non-metallic materials and ground planes or electromagnetic shielding can be used to reduce or further reduce electromagnetic interference (EMI).
[0041] In some examples, the temperature-sensing integrated circuit has a small form factor that is less than 3 mm in at least two of three orthogonal dimensions. In some examples, the temperature-sensing integrated circuit has a small form factor that is less than 2 mm in all three orthogonal dimensions. In some examples, the body of the sensor probe has an outer diameter that is less than or equal to 4 mm and an inner diameter that is less than or equal to 3 mm.
[0042] Referring now to FIGS. **1** and **2**, examples of plasma processing chambers that may use the sensor probes are shown. As can be appreciated, the sensor probes can be used in a variety of other types of semiconductor processing equipment such as cooled pedestals, spin chucks, processing chambers, etc. In FIG. **1**, an example of a substrate processing system **110** according to the present disclosure is shown. The substrate processing system **110** includes a processing chamber **122** that encloses other components of the substrate processing system **110** and contains the RF plasma (if used). The substrate processing system **110** includes an upper electrode **124** and a substrate support **126** such as an electrostatic chuck (ESC). During operation, a substrate **128** is arranged on the substrate support **126**.
[0043] For example only, the upper electrode **124** may include a gas distribution device **129** such as a showerhead that introduces and distributes process gases. The gas distribution device **129** may include a stem portion including one end connected to a top surface of the processing chamber. A base portion is generally cylindrical and extends radially outwardly from an opposite end of the stem portion at a location that is spaced from the top surface of the processing chamber. A substrate-facing surface or faceplate of the base portion of the showerhead includes a plurality of holes through which precursor, reactants, etch gases, inert gases, carrier gases, other process gases or purge gas flows. Alternately, the upper electrode **124** may include a conducting plate and the process gases may be introduced in another manner.
[0044] The substrate support **126** includes a baseplate **130** that acts as a lower electrode. The baseplate **130** supports a heating plate **132**, which may correspond to a ceramic multi-zone heating plate. A thermal resistance layer **134** may be arranged between the heating plate **132** and the baseplate **130**. The baseplate **130** may include one or more channels **136** for flowing coolant through the baseplate **130**.
[0045] An RF generating system **140** generates and outputs an RF voltage to one of the upper electrode **124** and the lower electrode (e.g., the baseplate **130** of the substrate support **126**). The other one of the upper electrode **124** and the baseplate **130** may be DC grounded, AC grounded or floating. For example only, the RF generating system **140** may include an RF generator **142** that generates RF plasma power that is fed by a matching and distribution network **144** to the upper electrode **124** or the baseplate **130**. In other examples, the plasma may be generated inductively or

remotely.
[0046] A gas delivery system **150** includes one or more gas sources **152-1**, **152-2**, . . . , and **152**-N (collectively gas sources **152**), where N is an integer greater than zero. The gas sources **152** are connected by valves **154-1**, **154-2**, . . . , and **154**-N (collectively valves **154**) and MFCs **156-1**, **156-2**, . . . , and **156**-N (collectively MFCs **156**) to a manifold **160**. Secondary valves may be used between the MFCs **156** and the manifold **160**. While a single gas delivery system **150** is shown, two or more gas delivery systems can be used.
[0047] A temperature controller **163** may be connected to a plurality of thermal control elements (TCEs) **164** arranged in the heating plate **132**. The temperature controller **163** may be used to control the plurality of TCEs **164** to control a temperature of the substrate support **126** and the substrate **128**. The temperature controller **163** may communicate with a coolant assembly **166** to control coolant flow through the channels **136**. For example, the coolant assembly **166** may include a coolant pump, a reservoir and/or one or more temperature sensors. The temperature controller **163** operates the coolant assembly **166** to selectively flow the coolant through the channels **136** to cool the substrate support **126**.
[0048] A valve **170** and pump **172** may be used to evacuate reactants from the processing chamber **122**. A system controller **180** may be used to control components of the substrate processing system **110**. The substrate processing system **210** may include a RF generating system **181** having an RF generator **182** and a matching network **184**. One or more sensor probes **190** may be inserted into cavities defined in the substrate support to sense a temperature of surface.
[0049] In FIG. **2**, another example of a substrate processing system **210** is shown. The substrate processing system **210** includes a coil driving circuit **211**. A pulsing circuit **214** may be used to pulse the RF power on and off or vary an amplitude or level of the RF power. The tuning circuit **213** may be directly connected to one or more inductive coils **216**. The tuning circuit **213** tunes an output of the RF source **212** to a desired frequency and/or a desired phase, matches an impedance of the coils **216** and splits power between the coils **216**. In some examples, the coil driving circuit **211** is replaced by one of the drive circuits described further below in conjunction with controlling the RF bias.
[0050] In some examples, a plenum **220** may be arranged between the coils **216** and a dielectric window **224** to control the temperature of the dielectric window **224** with hot and/or cold air flow. The dielectric window **224** is arranged along one side of a processing chamber **228**. The processing chamber **228** further comprises a substrate support (or pedestal) **232**. The substrate support **232** may include an electrostatic chuck (ESC), or a mechanical chuck or other type of chuck. Process gas is supplied to the processing chamber **228** and plasma **240** is generated inside of the processing chamber **228**. The plasma **240** etches an exposed surface of the substrate **234**. A drive circuit **252** (such as one of those described below) may be used to provide an RF bias to an electrode in the substrate support **232** during operation.
[0051] A gas delivery system **256** may be used to supply a process gas mixture to the processing chamber **228**. The gas delivery system **256** may include process and inert gas sources **257**, a gas metering system **258** such as valves and mass flow controllers, and a manifold **259**. A gas delivery system **260** may be used to deliver gas **262** via a valve **261** to the plenum **220**. The gas may include cooling gas (air) that is used to cool the coils **216** and the dielectric window **224**. A heater/cooler **264** may be used to heat/cool the substrate support **232** to a predetermined temperature. A temperature controller **263** may be included and connected to the controller **254**, the substrate support **232**, and the coolant assembly **264**. An exhaust system **265** includes a valve **266** and pump **267** to remove reactants from the processing chamber **228** by purging or evacuation.
[0052] A controller **254** may be used to control the etching process. The controller **254** monitors system parameters and controls delivery of the gas mixture, striking, maintaining and extinguishing the plasma, removal of reactants, supply of cooling gas, and so on. Additionally, as described below in detail, the controller **254** may control various aspects of the coil driving circuit **211** and the drive

circuit **252**. One or more sensor probes **190** may be inserted into cavities of the substrate support to sense a temperature a surface.

[0053] Referring now to FIGS. **3**, **4**A and **4**B, a substrate support **300** such as an electrostatic chuck (ESC) includes a baseplate **310** arranged adjacent to a heater layer **314**. While a baseplate of a substrate support is shown, the sensor probe can be used to sense the temperature of a surface of other components of substrate processing equipment. The heater layer **314** includes heaters **316**. A ceramic layer **318** including electrodes **320** is arranged adjacent to the heater layer **314**. A sensor probe **190** is inserted into cylindrical cavity **332**.

[0054] In some examples, the sensor probe **190** includes an elongated body **330** including a first end portion **334**. The first end portion **334** of the sensor probe **190** has a diameter that is greater than a diameter of an elongated body **330**. The elongated body **330** is received by a threaded housing **336** located at one end of the elongated body **330**.

[0055] The threaded housing **336** includes a first portion **338**, a second portion **340** and a third portion **344**. In some examples, the first portion **338**, the second portion **340** and the third portion **344** are cylindrical and include aligned internal cavities. The second portion **340** has a diameter that is greater than a diameter of the first portion **338**. The second portion **340** includes threads **346** that are received in threaded bores **348** in the baseplate **310** of the substrate support **300**. The third portion **344** projects radially outwardly from the baseplate **310** to allow the sensor probe **190** to be rotated relative to the baseplate **310** to allow insertion and removal.

[0056] A printed circuit board (PCB) **354** such as a flexible PCB passes through the elongated body **330** and extends from the third portion **344** of the threaded housing **336**. The PCB **354** is connected by a connector **356** (such as a PCB) to one or more wires **360** supplying power, ground and one or more signal lines to and from an integrated circuit located in the sensor probe **190**.

[0057] In FIG. **4**A, the elongated body **330** of the sensor probe **190** includes an inclined surface **361** which provides a transition from the diameter of the elongated body **330** to the larger diameter of the first end portion **334**. A spacer **362** projects from the elongated body **330** in a radial direction to evenly space the elongated body **330** within the cylindrical cavity **332**. An inclined surface **372** provides a transition from the first portion **338** to the second portion **340** of the threaded housing **336**. In some examples, an O-ring **376** is arranged against the inclined surface **372** and acts as an RF gasket.

[0058] In FIG. **4**B, the elongated body **330** includes a radially-projecting surface **408** located along the first end portion **334**. The sensor probe **190** includes a spring **410** located around the first end portion **334** adjacent to an inner surface of a cavity formed in the threaded housing **336**. The spring **410** biases an end of the elongated body **330** against a surface to determine a temperature of the surface.

[0059] A first integrated circuit **420** is mounted on the PCB **354**. The first integrated circuit **420** senses a first temperature of the surface to be monitored. In some examples, a second integrated circuit **422** is mounted on the PCB **354**. The second integrated circuit **422** senses a temperature that is remote from the surface to be monitored and is used to increase the reliability of the temperature measured by the first integrated circuit **420**, for diagnostic monitoring and/or for rationality checks of the first integrated circuit **420**.

[0060] In FIG. **5**, the PCB **354** is shown to include a first portion **508** that extends a distance greater than the elongated body **330** of the sensor probe **190**. The first integrated circuit **420** is mounted at one end of the first portion **508**. If used, the second integrated circuit **422** is mounted at a location spaced from the first integrated circuit **420**. A second end **510** of the PCB **354** includes terminals **526** connected to traces **522** (partially shown). The PCB **354** includes two or more layers including conductive traces, vias, ground planes, etc. to provide connections from the terminals **526** to the first integrated circuit **420** and/or the second integrated circuit **422**.

[0061] Referring now to FIG. **6**, the elongated body **330** may include one or more slots **610** to allow heat transfer. In some examples, the slots **610** have an elongated elliptical shape and are aligned in an axial direction of the elongated body **330**.

[0062] Referring now to FIG. **7**, a circuit **700** including the first integrated circuit **420**, resistors R**1** and R**2** and a capacitor C**1** is shown. A voltage supply line V.sup.+ is connected to a V.sup.+ terminal of the first integrated circuit **420**. A first reference potential such as ground is connected to a GND input and a second input of the first integrated circuit **420**. Signal lines S**1** and S**2** are connected to SDA and SGL lines of the first integrated circuit **420**. Resistors R**1** and R**2** are connected between the voltage supply line V.sup.+ and the signal lines S**1** and S**2**.

[0063] Referring now to FIG. **8**, attachment of the first integrated circuit **420** to the PCB **354** is shown. Solder bumps **810** connect pads on the first integrated circuit **420** to corresponding pads on the PCB **354**.

[0064] Referring now to FIGS. **9**A and **9**B, attachment of the first integrated circuit **420** and the PCB **354** to a cap **820** is shown. In some examples, the cap **820** is made of metal and includes legs **822** and **824** extending from one side thereof. In FIG. **9**A, the cap **820** is arranged or attached with the legs **822** and **824** extending transverse to a mounting surface of the PCB **354**. In FIG. **9**B, the cap **820** is arranged or attached with the legs **822** and **824** extending parallel to the mounting surface of the PCB **354**.

[0065] Referring now to FIGS. **10**A and **10**B, potting of the first integrated circuit **420** to the cap **820** is shown. Potting material **1010** is applied to attach the cap **820** to the integrated circuit.

[0066] Referring now to FIGS. **11**A and **11**B, the PCB **354**, the first integrated circuit **420**, and the cap **820** are inserted into a cavity **1100** of the elongated body **330**. In FIG. **11**B, the PCB **354** is bent at a right angle to allow the legs **822** and **824** of the cap **820** to be inserted into an end of the elongated body **330**.

[0067] Referring now to FIGS. **12** and **13**, another example of the sensor probe **1200** is shown. In FIG. **12**, the sensor probe **1200** includes an elongated body **1210** connected to a threaded housing **1214**. In some examples, the threaded housing **1214** includes a threaded surface **1216**.

[0068] In FIG. **13**, a first integrated circuit **1320** is mounted by solder balls **1330** to a first PCB **1326**. A capacitor **1334** is mounted on the first PCB **1326**. A second PCB **1350** is arranged in the threaded housing **1214** and includes a resistor **1360** mounted thereto. One or more wires **1362** provide external connections. One or more wires or hard PCB traces **1366** provide connections between the first PCB **1326** and the second PCB **1350**. Potting material **1370** is located inside of the elongated body **1210**. The first integrated circuit **1320** senses a temperature of a surface **1380**.

[0069] Referring now to FIG. **14**, if additional shielding is desired, shielding layers **1410** can be used to cover upper and/or lower surfaces of a PCB **1400** to provide enhanced shielding. In some examples, the shielding layers **1410** include a metal layer. In some examples, the shielding layers **1410** are connected to a reference potential such as ground. In other examples, the shielding layer **1410** includes a plurality of conductors. In some examples, the plurality of conductors are uniformly spaced and form a grid in one or more transverse directions. The plurality of conductors are connected to a reference potential such as ground.

[0070] Referring now to FIG. **15**, the sensor probes can also be implemented without using printed circuit boards. While two sensor probes are shown to show variations in the position of the temperature-sensing integrated circuit, one or more than two sensor probes can be used in a given application. First and second sensor probes **1510-1** and **1510-2** include integrated circuits **1520-1** and **1520-2** that are located in the elongated bodies **1210-1** and **1210-2**, respectively, adjacent to a surface-facing end of the elongated bodies **1210-1** and **1210-2**, respectively. A plurality of solder balls **1530-1** and **1530-2** provide connections to the integrated circuits **1520-1** and **1520-2**, respectively. A plurality of wires **1562** are soldered to selected ones of the plurality of solder balls **1530-1** and **1530-2** to provide one or more external connections to the integrated circuits **1520-1** and **1520-2** through the threaded housings **1214-1** and **1214-2** and the elongated bodies **1210-1** and **1210-2**, respectively. In some examples, the wires **1562** include insulated conductors.

[0071] Potting material **1570-1** and **1570-2** is located inside of the elongated bodies **1210-1** and

1210-2, respectively. The integrated circuits **1520-1** and **1520-2** sense a temperature of a surface **1580**. The integrated circuits **1520-1** and **1520-2** can be arranged parallel to the elongated bodies **1210-1** and **1210-2**, perpendicular to the elongated bodies **1210-1** and **1210-2**, or at angles therebetween.

[0072] In some examples, the number of solder balls S is equal to the number of wires W, where S and W are integers greater than one. In other examples, S>W or W>S.

[0073] The foregoing description is merely illustrative in nature and is in no way intended to limit the disclosure, its application, or uses. The broad teachings of the disclosure can be implemented in a variety of forms. Therefore, while this disclosure includes particular examples, the true scope of the disclosure should not be so limited since other modifications will become apparent upon a study of the drawings, the specification, and the following claims. It should be understood that one or more steps within a method may be executed in different order (or concurrently) without altering the principles of the present disclosure. Further, although each of the embodiments is described above as having certain features, any one or more of those features described with respect to any embodiment of the disclosure can be implemented in and/or combined with features of any of the other embodiments, even if that combination is not explicitly described. In other words, the described embodiments are not mutually exclusive, and permutations of one or more embodiments with one another remain within the scope of this disclosure.

[0074] Spatial and functional relationships between elements (for example, between modules, circuit elements, semiconductor layers, etc.) are described using various terms, including “connected,” “engaged,” “coupled,” “adjacent,” “next to,” “on top of,” “above,” “below,” and “disposed.” Unless explicitly described as being “direct,” when a relationship between first and second elements is described in the above disclosure, that relationship can be a direct relationship where no other intervening elements are present between the first and second elements, but can also be an indirect relationship where one or more intervening elements are present (either spatially or functionally) between the first and second elements. As used herein, the phrase at least one of A, B, and C should be construed to mean a logical (A OR B OR C), using a non-exclusive logical OR, and should not be construed to mean “at least one of A, at least one of B, and at least one of C.”

[0075] In some implementations, a controller is part of a system, which may be part of the above-described examples. Such systems can comprise semiconductor processing equipment, including a processing tool or tools, chamber or chambers, a platform or platforms for processing, and/or specific processing components (a wafer pedestal, a gas flow system, etc.). These systems may be integrated with electronics for controlling their operation before, during, and after processing of a semiconductor wafer or substrate. The electronics may be referred to as the “controller,” which may control various components or subparts of the system or systems. The controller, depending on the processing requirements and/or the type of system, may be programmed to control any of the processes disclosed herein, including the delivery of processing gases, temperature settings (e.g., heating and/or cooling), pressure settings, vacuum settings, power settings, radio frequency (RF) generator settings, RF matching circuit settings, frequency settings, flow rate settings, fluid delivery settings, positional and operation settings, wafer transfers into and out of a tool and other transfer tools and/or load locks connected to or interfaced with a specific system.

[0076] Broadly speaking, the controller may be defined as electronics having various integrated circuits, logic, memory, and/or software that receive instructions, issue instructions, control operation, enable cleaning operations, enable endpoint measurements, and the like. The integrated circuits may include chips in the form of firmware that store program instructions, digital signal processors (DSPs), chips defined as application specific integrated circuits (ASICs), and/or one or more microprocessors, or microcontrollers that execute program instructions (e.g., software). Program instructions may be instructions communicated to the controller in the form of various individual settings (or program files), defining operational parameters for carrying out a particular process on or for a semiconductor wafer or to a system. The operational parameters may, in some embodiments, be part of a recipe defined by process engineers to accomplish one or more processing steps during the fabrication of one or more layers, materials, metals, oxides, silicon, silicon dioxide, surfaces, circuits, and/or dies of a wafer.

[0077] The controller, in some implementations, may be a part of or coupled to a computer that is integrated with the system, coupled to the system, otherwise networked to the system, or a combination thereof. For example, the controller may be in the "cloud" or all or a part of a fab host computer system, which can allow for remote access of the wafer processing. The computer may enable remote access to the system to monitor current progress of fabrication operations, examine a history of past fabrication operations, examine trends or performance metrics from a plurality of fabrication operations, to change parameters of current processing, to set processing steps to follow a current processing, or to start a new process. In some examples, a remote computer (e.g. a server) can provide process recipes to a system over a network, which may include a local network or the Internet. The remote computer may include a user interface that enables entry or programming of parameters and/or settings, which are then communicated to the system from the remote computer. In some examples, the controller receives instructions in the form of data, which specify parameters for each of the processing steps to be performed during one or more operations. It should be understood that the parameters may be specific to the type of process to be performed and the type of tool that the controller is configured to interface with or control. Thus as described above, the controller may be distributed, such as by comprising one or more discrete controllers that are networked together and working towards a common purpose, such as the processes and controls described herein. An example of a distributed controller for such purposes would be one or more integrated circuits on a chamber in communication with one or more integrated circuits located remotely (such as at the platform level or as part of a remote computer) that combine to control a process on the chamber.

[0078] Without limitation, example systems may include a plasma etch chamber or module, a deposition chamber or module, a spin-rinse chamber or module, a metal plating chamber or module, a clean chamber or module, a bevel edge etch chamber or module, a physical vapor deposition (PVD) chamber or module, a chemical vapor deposition (CVD) chamber or module, an atomic layer deposition (ALD) chamber or module, an atomic layer etch (ALE) chamber or module, an ion implantation chamber or module, a track chamber or module, and any other semiconductor processing systems that may be associated or used in the fabrication and/or manufacturing of semiconductor wafers.

[0079] As noted above, depending on the process step or steps to be performed by the tool, the controller might communicate with one or more of other tool circuits or modules, other tool components, cluster tools, other tool interfaces, adjacent tools, neighboring tools, tools located throughout a factory, a main computer, another controller, or tools used in material transport that bring containers of wafers to and from tool locations and/or load ports in a semiconductor manufacturing factory.

Claims

1. A sensor probe, comprising: an elongated body defining an inner cavity having an inner diameter; a first printed circuit board configured to be fitted within the inner cavity; a temperature-sensing integrated circuit mounted on the first printed circuit board; a housing configured to receive one end of the elongated body and configured to be mounted to a baseplate of a substrate support; a second printed circuit board arranged in the housing; a plurality of first conductors connecting the first printed circuit board to the second printed circuit board; and a plurality of second conductors configured to connect the second printed circuit board to external devices.

2. The sensor probe of claim 1, wherein the first printed circuit board has a width that is less than the inner diameter and a length that is less than a length of the elongated body.

3. The sensor probe of claim 1, wherein the second printed circuit board has a length that is less than a length of the housing.
4. The sensor probe of claim 1, wherein the inner diameter is less than or equal to 3 mm and wherein at least two of three orthogonal dimensions of the temperature-sensing integrated circuit are less than 3 mm.
5. The sensor probe of claim 1, further comprising potting material located inside of the elongated body.
6. The sensor probe of claim 1, wherein the first printed circuit board and the temperature-sensing integrated circuit are mounted parallel to a length of the elongated body.
7. The sensor probe of claim 1, wherein the temperature-sensing integrated circuit senses a temperature of a surface in contact therewith.
8. The sensor probe of claim 7, wherein the surface is a layer within an electrostatic chuck.
9. The sensor probe of claim 1, wherein the elongated body includes a radial projection to center the elongated body in a cavity of the baseplate.
10. The sensor probe of claim 1, wherein the elongated body includes a slot.
11. The sensor probe of claim 10, wherein the slot has an elongated elliptical shape and is aligned in an axial direction of the elongated body.
12. The sensor probe of claim 1, further comprising a capacitor connected to the first printed circuit board.
13. The sensor probe of claim 1, further comprising a resistor connected to the second printed circuit board.
14. The sensor probe of claim 1, further comprising a shielding layer arranged on a surface of the first printed circuit board.
15. The sensor probe of claim 1, wherein: the first printed circuit board is disposed within the inner cavity and extends to a first end of the elongated body; and the temperature-sensing integrated circuit is mounted at a first end of the first printed circuit board and disposed within and at the first end of the elongated body.
16. The sensor probe of claim 1, wherein the temperature-sensing integrated circuit comprises solder balls, which are attached to the first printed circuit board.
17. The sensor probe of claim 1, wherein: the temperature-sensing integrated circuit measures a temperature of the baseplate; and the housing is configured to be mounted to a bottom of the baseplate and extend at least partially into the baseplate.
18. The sensor probe of claim 1, wherein the housing screws into the baseplate.
19. The sensor probe of claim 1, wherein the housing and the elongated body are configured to be mounted at least partially within the baseplate of the substrate support.
20. A sensor probe, comprising: an elongated body defining an inner cavity having an inner diameter; a temperature-sensing integrated circuit configured to be fitted within the inner cavity; a housing configured to receive one end of the elongated body and configured to be mounted to a baseplate of a substrate support; and a plurality of conductors configured to pass through the housing and the elongated body and to connect the temperature-sensing integrated circuit to external devices.
21. The sensor probe of claim 20, wherein the inner diameter is less than or equal to 3 mm and wherein at least two of three orthogonal dimensions of the temperature-sensing integrated circuit are less than 3 mm.
22. The sensor probe of claim 21, further comprising potting material located inside of the elongated body.
23. The sensor probe of claim 21, wherein the temperature-sensing integrated circuit is mounted parallel to a length of the elongated body.
24. The sensor probe of claim 21, wherein the temperature-sensing integrated circuit is mounted perpendicular to a length of the elongated body.

25. The sensor probe of claim 21, wherein the temperature-sensing integrated circuit senses a temperature of a surface in contact therewith.
26. The sensor probe of claim 25, wherein the surface is a layer within an electrostatic chuck.
27. The sensor probe of claim 21, wherein the elongated body includes a radial projection to center the elongated body in a cavity of the baseplate.
28. The sensor probe of claim 21, wherein the elongated body includes a slot.
29. The sensor probe of claim 28, wherein the slot has an elongated elliptical shape and is aligned in an axial direction of the elongated body.
30. The sensor probe of claim 21, further comprising a plurality of solder balls attached to the temperature-sensing integrated circuit, wherein the plurality of conductors are attached to the solder balls.
31. The sensor probe of claim 21, further comprising a printed circuit board disposed within the inner cavity and extending to a first end of the elongated body, wherein the temperature-sensing integrated circuit is mounted at the first end of the printed circuit board and disposed within and at the first end of the elongated body.
32. The sensor probe of claim 31, wherein the temperature-sensing integrated circuit comprises solder balls, which are attached to the printed circuit board.
33. The sensor probe of claim 21, wherein: the temperature-sensing integrated circuit measures a temperature of the baseplate; and the housing is configured to be mounted to a bottom of the baseplate and extend at least partially into the baseplate.
34. The sensor probe of claim 21, wherein the housing screws into the baseplate.
35. The sensor probe of claim 21, wherein the housing and the elongated body are configured to be mounted at least partially within the baseplate of the substrate support.
